

# Machine Learning Answers (~50% Correct)

## Q1:

Explain the concept of backpropagation in neural networks. (10 Marks)

**Answer:**

Backpropagation is a training algorithm used in neural networks to update weights. It calculates the error between predicted and actual outputs and propagates this error backward. Gradients are computed using the chain rule and weights are updated to reduce error. However, the explanation does not fully describe the mathematical update process.

## Q2:

What is overfitting in machine learning? How can it be prevented? (5 Marks)

**Answer:**

Overfitting occurs when a model learns the training data too well including noise. This causes poor performance on unseen data. It can be reduced using regularization, dropout, or cross-validation, although this explanation does not describe how each technique works in detail.

## Q3:

Define gradient descent and explain its role in training neural networks. (5 Marks)

**Answer:**

Gradient descent is an optimization algorithm used to minimize the loss function. It updates model weights in the direction of the negative gradient. This helps reduce prediction error over time. However, the explanation does not include learning rate tuning or different gradient descent types.

## Q4:

What is the difference between classification and regression in machine learning? (5 Marks)

**Answer:**

Classification and regression are supervised learning tasks. Classification predicts discrete categories such as spam or not spam, while regression predicts continuous values such as house prices. This answer does not explain specific algorithms used in each case.